Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Adhiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

Mid Term (Odd) Semester Examination October 2024

Roll No.....

Name of the Program: B.Tech ECE Semester: III
Name of the Course: Electronics & Communication Engineering
Course Code: TEC-302
Paper Name: Digital System Design

Time: 1:30 Hours **MM: 50**

Note:
(i) Answer **all the questions** by choosing **any one of the sub questions**.
(ii) Each question contains two parts a & b. Attempt any one part of choice from each question.

Q.1	**(10 Marks)**	CO1
a)	i. $(342.55)_{10} = (?)_8$ ii. $(381B)_{16} = (?)_2$ iii. $(101101.001)_2 = (?)_{10}$ iv. $(1111101)_2 = (\quad)_{16}$ v. $(101111011)_2 = (\quad)_8$	
	OR	
b)	Simplify the given boolean expression F(A,B,C)=A′B+BC′+BC+AB′C′ F(A,B,C)=(A+B)(A+C)	
Q.2	**(10 Marks)**	CO2
a)	Explain even and odd parity codes? Generate seven-bit Hamming code for the message bits of "0101" using code pattern from right to left.	
	OR	
b)	Explain SOP and POS terms. Simplify the following using K-Map. $f(w,x,y,z) = \sum(1,3,7,9,11,12,14) + \sum_d(2,5,6)$	
Q.3	**(10 Marks)**	CO2
a)	Minimize the function using k-map and realize it using NAND universal gates $F(P, Q, R, S) = \sum m\,(3,4,5,7,9,13,14,15) + \sum_{\varphi}(0,2,8)$	
	OR	
b)	Explain Octal, Excess-3 and Gray Number system. Subtract the following using 2's complement method (i) (+39) - (+16) (ii) (+49) - (+32)	
Q.4	**(10 Marks)**	CO1
a)	Explain Boolean laws. Express the Boolean function F = A + B'C as standard sum of products.	

Mid Term (Odd) Semester Examination October 2024

	Express the Boolean function $F = XY + X'Z$ as a standard product of sum.	
	OR	
b)	Realize $Y = (\bar{C}A + B.(A + CD))$ using NAND gates. $Y = (A + B)(C + B)(A + C)$ using NOR gates	
Q.5	**(10 Marks)**	CO2
a)	Using K map method simplify (i). $f(w, x, y, z) = \sum(1,3,4,5,6,7,9,12,13)$ (ii). $F(A, B, C, D) = \prod(0,1,4,5,6,8,9,12,13,14)$	
	OR	
b)	What do you understand by K-Map simplification? Explain four variable K Map. Find the minimal expression using K Map: $$F(A, B, C, D) = \sum(1,3,7,11,15) + \sum_{\phi}(0,2,5)$$	